

SciOdyssey.

A research layer over your agent harness.

PRESERVE THE RESEARCH WORLD · RESET THE RESEARCHER

ROBUST

SUPERVISABLE

BROAD SEARCH

SAME EVIDENCE
FRESH JUDGMENT
A WIDER SEARCH

Research
Runs
Longer.

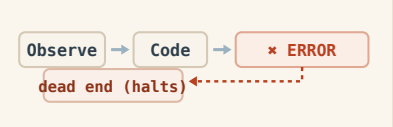
01 / PROBLEM

Why current agents fail.

Sustained autonomous research breaks in three different ways.

01

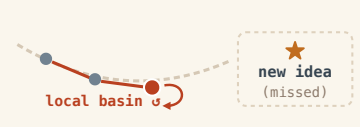
Closed-world fragility
Fixed workflows shatter on unexpected runtime errors.



Unhandled runtime faults halt unassisted runs

02

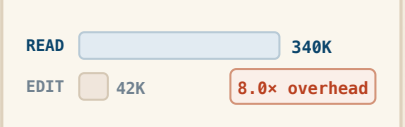
Trajectory momentum
A single context carries past momentum into local basins.



22 runs trapped tweaking parameters in same basin

03

Context inflation cost
Reconstructing context costs far more than the edit itself.



10 files read to change 1 line; burns token budget

DESIGN RESPONSE

Keep the world. Reset the trajectory.

Decouple persistent memory from reasoning

1

Auto-heal runtime faults

2

Fresh reasoning each cycle

3

Reusable disk state (no token waste)

02 / SYSTEM

One persistent world. Fresh scientific judgment.

Give the research world and the researcher different lifetimes.

DUAL-LIFETIME LOOP

Persistent world · ephemeral trajectory · audited write-back

INPUTS

TASK

DATA

EVALUATOR

BUDGET

Fresh Scientist

Owns scientific judgment inside one trajectory: hypotheses, code, experiments, interpretation, and pivots.

META Review

Audits claims against artifacts, scopes conclusions, and decides what is durable enough to survive.

SELECTIVE PERSISTENCE

Only audited empirical findings, code state, and bounded priors write back to the shared world.

PERSISTENT RESEARCH WORLD

What survives across trajectories

A · EVIDENCE

Knowledge + Ledger

Metrics, failures, reports, and experiment receipts in `ledger.jsonl`.

B · PRIORS

Brief + Hypotheses

Current understanding and exploration leads in `brief.md`, open to revision.

C · STATE

Code + State

Retained implementation, serialized model weights, checks, and git provenance.

CONTEXT RESET

The next Scientist reads verified experience from disk, but does not inherit the previous reasoning trace.

RUNTIME SUBSTRATE

WORKTREE ISOLATION

Independent git worktrees isolate branches and preserve reproducible artifacts.

SELF-HEALING

Traceback → patch → rerun keeps autonomous research moving after runtime faults.

PARALLEL SYNTHESIS

Completed branches can be compared post-hoc; useful ideas survive even when a branch loses.

The next Scientist inherits **verified experience** — not prior reasoning momentum.

Decoupled Memory · Audited Persistence · Fresh Scientific Judgment

03 / EVALUATION

Five empirical claims. Measured evidence.

Standardized evaluation on KuaiRand-Pure benchmark · Fixed official evaluator · Complete telemetry accounting

01 MEASURED RESULT

The model got better.

0.6016000 → 0.6059363

Official FM baseline SciOdyssey submission

+0.0043363 Primary improvement

GAUC 0.6728421 (+0.0054)

nDCG@5 0.5390304 (+0.0033)

LogLoss 0.38102 (-0.0032)

Feature Engine: 46 categorical cross-interactions + log1p transforms

Architecture: 8-seed Factorization Machine ensemble (NumPy / CPU)

Monotonic gain from baseline (0.6016) → feature engineering (0.6041) → seed ensembling (0.6059); 100% CPU inference without GPU clusters.

03 SELF-HEALING & GATEKEEPING

The agent recovered from failure.

TIER 1 · RUNTIME SELF-HEALING

CRASH AUTO-RECOVERY

FAILURE Cycle 1 `numpy.float32` serialization crash

ACTION Scientist parsed traceback, auto-patched scalar helper

OUTCOME 0 human bugfixes · Unattended rerun succeeded

TIER 2 · META GATEKEEPING

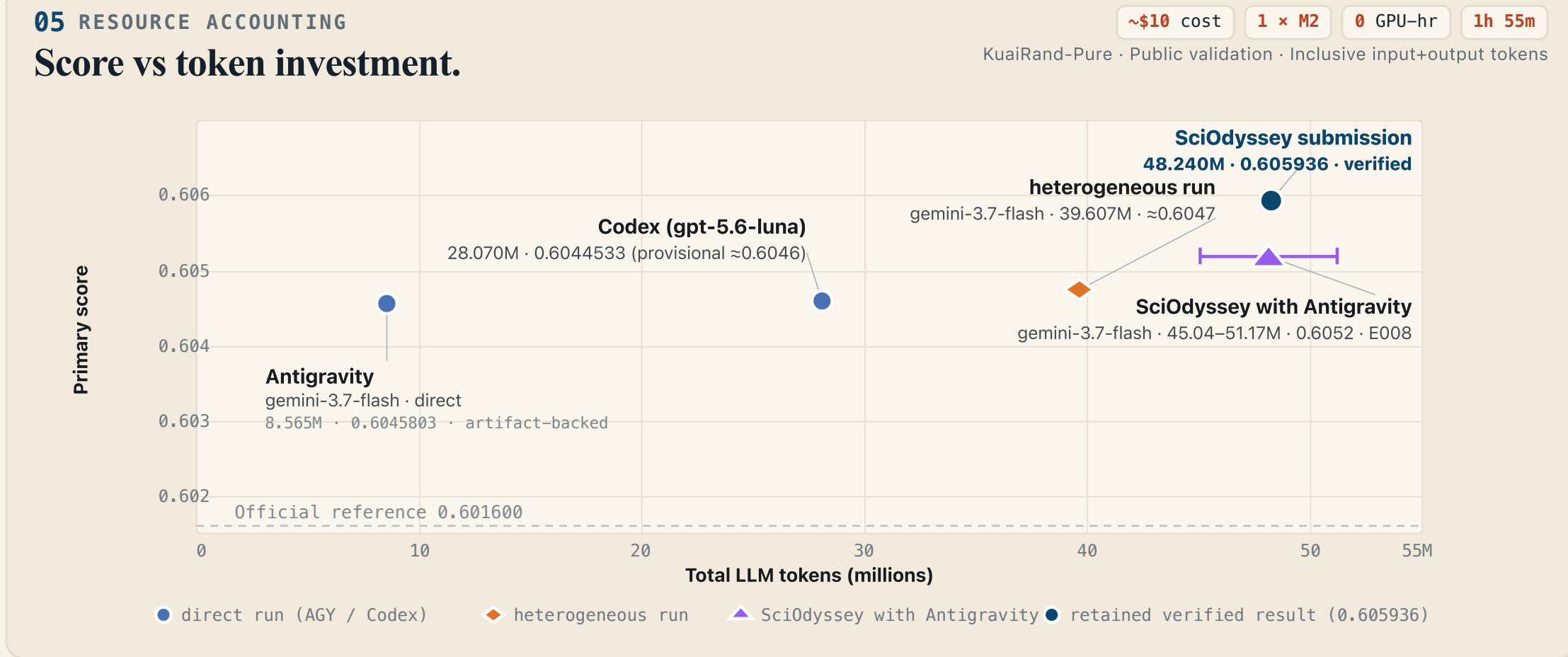
SPURIOUS STATE REJECTION

FALSE CLAIM Scientist claimed BPR gain on medium proxy

ACTION META audited `system/data.py`, caught `UNK` leak

OUTCOME Promotion blocked · Zero contaminated state

Empirical receipts preserved: runtime self-healing telemetry and independent code-level proxy audits.



All results **verified** on KuaiRand-Pure official validation split with **complete reproducible artifacts**.

04 / FINDINGS

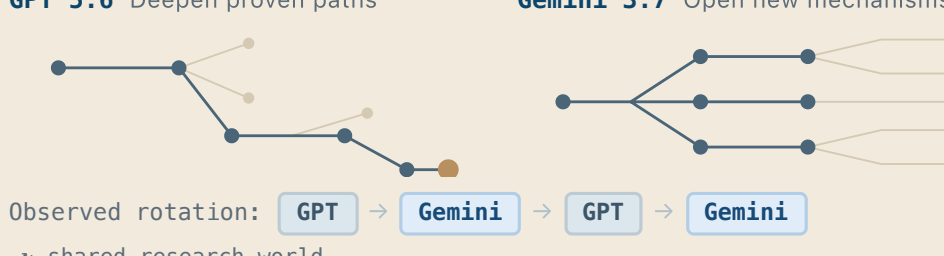
What the run revealed.

01 · MODEL DIVERSITY

Model diversity can help break plateaus.

GPT 5.6 Deepen proven paths

Gemini 3.7 Open new mechanisms



Gemini-only 0.6052 → GPT+Gemini 0.6059

02 · PARALLEL SYNTHESIS

A losing branch can still improve the winner.

A 0.6048 Redundant

B 0.6048 Chosen ✓

C 0.6039 Useful ideas

Reviewer Synthesize B+C → Fresh Scientist 0.6055536

Selection keeps useful knowledge, not just the top score: Branch C contributed pairwise ranking and rank normalization.

Branch B+C Ideas → Synthesis 0.6055536

05 / EXPERIENCE

Start your journey.

task.md

What to solve

Formal task contract: research objectives, dataset inputs, compute constraints, and official evaluation protocol.

Dataset paths & split integrity

Official sandbox evaluator

PERSONAL.md

How to work

User research preferences, model guidance, exploration style, tool permissions, and supervisory checkpoints.

Model rotation (GPT / Gemini)

Supervision & budget thresholds

01 Contract

task.md

→

02 Probe

step

→

03 Search

run

→

04 Branch

parallel

→

05 Audit

gui

SUPERVISION TIERS

Human in the loop · Autonomous on demand

1 Unattended Runs

Overnight budget execution; auto-heals runtime crashes without manual intervention.

2 Checkpoint Review

META audits empirical artifacts before durable write-back to the project world.

3 Steer Working Priors

Adjust PERSONAL.md or inject domain hypotheses between fresh trajectories.

DECLARATIVE PROTOCOL

Contract boundary: Human defines objectives & preferences; SciOdyssey autonomously iterates, verifies, and records evidence.

Autonomous by default.

Supervisable by design.

Coding agent harness · Execution studio

Works with: Codex Claude Code Antigravity OpenCode Trae

> SKILL.md "Use the research-agent skill. Explore in parallel; let me review what to keep."

\$ research-agent step # one cycle · human reviews

META → Scientist → Evidence → Your turn ✓

\$ research-agent run --max-cycles 4 # autonomous · fresh scientist each round

C1 → C2 → C3 → C4 · Candidate Retained ✓

\$ research-agent parallel --branches 3 # isolated worktrees · reviewer synthesizes

Branch A·B·C → Reviewer Synthesis → Explicit Promotion ✓

\$ research-agent gui # dashboard · telemetry & checkpoints

Telemetry · Loss Curves · Verified Checkpoints ✓

